

IPOs, trade sales and liquidations: Modelling venture capital exits using survival analysis

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Abstract

This paper examines the dynamics of exit options for US venture capital funds. Using a sample of more than 20,000 investment rounds, we analyze the time to ‘IPO’, ‘trade sale’ and ‘liquidation’ for 6000 VC-backed firms. We model these exit times using competing risks models, which allow for a joint analysis of exit type and exit timing. The hazard rate for IPOs are clearly non-monotonic with respect to time. As time flows, VC-backed firms first exhibit an increased likelihood of exiting to an IPO. However, after having reached a plateau, non-exited investments have fewer possibilities of IPO exits as time increases. This sharply contrasts with trade sale exits, where the hazard rate is less time-varying. We further provide evidence on the impact of economic factors such as syndicate size and composition, geographical location and VC value adding, on exit outcomes.

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1. Introduction

The assessment of possible exit options is of paramount importance for venture capitalists prior to their investments in new ventures. Indeed, not only are they concerned about *how* they can cash out but also *how long* they need to be involved in their portfolio companies before cashing out. Besides the offer price, the exit decision therefore features two important dimensions. First, the type of exit route and secondly the actual timing of the exit.³ The existing academic literature on venture capital exits has shown that venture capitalists time their exits using stage financing split into several rounds (e.g., Gompers, 1995; Bergemann and Hege, 1998; Cornelli and Yosha, 2003). Besides the disciplinary action that this procedure exerts on venture capital-backed firms, this stage financing also provides exit options to venture capitalists at all financing rounds. In addition to stage finance, several studies document the widespread use of contractual arrangements that guarantee the venture capitalist explicit intervention rights, also regarding exit issues (Gompers, 1997; Cumming, 2002; Kaplan and Stromberg, 2003). More generally, these rights allow the venture capitalist to force an exit.

While IPO exits by VC funds have been researched quite extensively,⁴ a stock market listing is however only one out of several ways to exit private equity investments. Interestingly, the academic literature has not focused much on other types of exits such as trade sales and liquidations. Nor do we know how these various exit routes interact as time flows. In this paper, we consider both dimensions (i.e., type and timing) of exit within a single framework of analysis. Thus, we explore the full range of exit routes, i.e., not only IPOs, but also trades sales and liquidations. This is particularly important when tackling the issue of ‘exit risk’ for venture capital investments as it requires jointly taking into account all potential exit routes as well as the duration until shares become liquid. We also assess how exit conditions evolve when firms move up the ladder of financing rounds and compare these results with the prevailing conditions at the initial (first round) investment. This provides a better picture of the dynamics of exit options for venture capital funds. We also tackle the issue as to whether the geographical location of the entrepreneurial firm affects the two dimensions of exit. For instance, does an investment in the Silicon Valley facilitate VC funds exits more than a similar investment in Texas or on Route 128?

In the empirical application, we analyze the time-to-exit through IPO, trade sale and liquidation for about 6000 venture-backed firms using a detailed sample of more than 20,000 investment rounds. In the framework of survival analysis, we characterize and model the times-to-exit using competing risks models, which allow for a joint analysis of exit type and exit timing as well as their dynamic interplay. To our knowledge, this is the first application of such statistical models to venture capital investments.⁵ The

³ Note that we also use the word ‘trade sales’ for so-called acquisitions.

⁴ See, among others, Lerner (1994), Gompers (1995, 1996), Black and Gilson (1998), Bascha and Walz (2001), Cumming (2002), Cumming and MacIntosh (2003), Das et al. (2003), Cochrane (2005), and Schwienbacher (2005).

⁵ Note that Gompers (1995) and Cumming and MacIntosh (2001) also use duration models, but these are not competing risks models. The strength of our approach lies in the rigorous statistical modelling of exit times, and the possibility to fully parameterize the exit times with explicative variables known at the time the investment round took place. The use of the generalized Gamma density distribution in the competing risks models allows for non-monotonic increasing/decreasing conditional probabilities of exits (also called hazard functions). For example, this allows for increasing (as time goes by) IPO hazard rates during the first n years, and thereafter decreasing conditional probabilities of exit.

empirical analysis delivers key results which can be summarized as follows. First, regarding the shape of the conditional probability (hazard rate) of IPO exit, a first sharply increasing hazard rate and then a decreasing hazard rate are observed. Thus, as time flows, venture capital-backed firms first exhibit an increased likelihood of exiting to an IPO. However, after having reached a plateau, investments that have not yet exited have fewer and fewer possibilities of IPO exits as time increases. We interpret this as evidence that IPO candidates tend to be selected relatively quickly. In contrast, if they do not achieve a public listing fast enough, their chances of doing so quickly decrease. For trade sales, hazard functions reach their maximum much later and tend to decrease slowly thereafter. This is very much in line with the widespread notion that, in contrast to an IPO, a trade sale is a more universal exit channel.

Second, the achievement of milestones in past rounds accelerates exit by all means. Achieving intermediate milestones leads to the resolution of uncertainty about whether the projects of a venture are likely to be profitable or whether they should be stopped. While the result on IPOs and trade sales is quite intuitive, the fact that it also accelerates liquidations is more puzzling. The degree of value-adding by VC funds also impacts exits. We find that a larger syndicate size accelerates exit of all types but the greatest impact occurs for IPOs. Proximity of at least one VC fund makes trade sales more likely, suggesting that greater control by VC funds is related to more acquisitions. Trade sales are significantly more likely for firms based in California (Silicon Valley) and northeast states (Route 128). Similarly, liquidations are less likely in these regions. Since these are also the regions that feature the largest clustering of entrepreneurial firms and venture capitalists (and all other important players involved in venture capital finance), this may suggest that such a concentration facilitates the success of these firms as compared to firms in other US regions. Finally, we look at the impact of stock market conditions on the exit dynamics and find that the exit of investments initiated at times of more favorable IPO conditions tended to be sped up as venture capitalists were probably eager to capitalize on better exit chances.

Our results thus shed light on the competing exit possibilities for venture capitalists and on the dynamics of the time-to-exit for the IPO, trade sale and liquidation exits. We therefore provide a broader perspective into the Black and Gilson's hypothesis on the link between the venture capital market and the stock market (Black and Gilson, 1998). It argues that an active IPO market facilitates VC exits and therefore makes VC investments more attractive. While this hypothesis is directly related to IPOs, our results extend to other exit routes. Our results also provide evidence that more liquid stock markets facilitate the *recycling of informed capital* (Michelacci and Suarez, 2004) in that it allows venture capitalists to redirect their human capital to new projects more quickly.

The rest of the paper is structured as follows. After this introduction, we provide a theoretical discussion of our relevant variables. In Section 3, we then detail the data and variables used in the analysis. We next present the competing risks model in Section 4. The empirical application is split into two sections: Section 5 gives a detailed descriptive analysis, while we present all the estimation results in Section 6. Finally, Section 7 concludes.

2. Exit decisions and type of exits for venture capital-backed firms

Our econometric modelling sets the stage for the exploration of two separate sets of dimensions of venture capital exits (expected time-to-exit and the type of exit). This

methodology provides a more accurate estimation of the considered effects on exit as compared to binary hazard models. Our framework also allows to capture changes in these probabilities over time. Furthermore, as will become clear below, most of our variables affect both dimensions in the same way. In the remainder of this section, we provide empirical predictions regarding a number of investment and VC fund characteristics for which we want to explore their effect on at least one of the two dimensions of exit.

2.1. Geographical location of the entrepreneurial company

Being located in a well-developed cluster of entrepreneurial activities and start-up finance may facilitate the success of ventures and exits of participating VC funds. In the US, the greatest concentration of venture capital activities is located in Silicon Valley and around Route 128. This may accelerate the cycle of business development as it is easier to find the needed resources and develop business contacts. Such an environment typically provides many other services needed for developing companies from scratch (see [Hellmann, 2001](#), for a related discussion on Silicon Valley). A recent study provides evidence that such network connections lead to better performance in venture capital ([Ljungqvist et al., forthcoming](#)). In addition, this effect is further strengthened by the clustering of entrepreneurial firms with similar objectives and entrepreneurial spirit in these same regions as it reduces informational costs (and other costs) of investors ([Michelacci and Suarez, 2004](#)). As pointed out by [Hellmann and Perotti \(2004\)](#), the circulation of novel ideas can be beneficial for start-up companies. Overall, we therefore expect a positive effect for companies located in these two regions on successful exits (mainly IPOs) and a negative effect on liquidation.

2.2. Regional proximity

Being located ‘not too far away’ from the investee enables the investor to better assist and monitor the entrepreneurial firm. This ought to increase the chance of success, possibly also speeding up exits as the venture capitalist may more easily (at less cost) monitor the investee. This provides investors with more relevant information and thus help them make better decisions as to extend finance or stop the investment. A recent study by [Cumming and Johan \(2005\)](#) shows that this affects the type of investments that VC funds are likely to undertake. To our knowledge, the impact of proximity on the ultimate exit of VC funds has not been analyzed empirically yet.

2.3. Technological improvement (achievement of milestones)

Companies in which venture capital firms invest are typically characterized by a great amount of asymmetric information, in particular which respect to the quality of innovation. A technological advancement can therefore be seen as a sign of quality. More specifically, a company that very quickly progresses in terms of stage of development would be a more successful firm. In the same vein, realizing technological progress from one round to the other signals the achievement of important ‘technological milestones’ (see, e.g., [Bergemann and Hege, 2002](#)). This in turn may accelerate its successful exit. We therefore expect

a positive (negative) effect on IPO (liquidation) likelihood.⁶ Note that the characterization of milestones is not limited to technological issues only. It may also include progress in team building, market research and other criteria that are typically considered as milestones of a firm's development.⁷ Clearly, achieving a milestone may represent getting better information about the product itself, beyond the technological dimension (see Berk et al., 2004, for a theoretical discussion in this differentiated view on various sources of idiosyncratic and systematic risk).

2.4. *Syndicate size*

Most venture capital deals are syndicated and the size of the syndicate tends to become larger as the venture gets more developed and requires greater amounts of capital. Many rationales have been suggested to explain syndication in venture capital deals (Barry et al., 1990; Megginson and Weiss, 1991; Admati and Pfleiderer, 1994; Lerner, 1994; Hellmann, 2002; Brander et al., 2002). Some of these studies suggest that larger syndicates should make exit easier for successful start-ups as far as it increases the pool of contacts required to make a trade sale possible. Having well-established venture capitalists in the syndicate may further facilitate the IPO process through enhanced certification, which may lead to less underpricing. Moreover, one can expect increased performance through greater complementarities of skills between participating syndicate members. We therefore expect that a larger syndicate increases the likelihood of exiting from a successful venture, either through a trade sale or an IPO.

2.5. *VC firm experience*

Similarly, an IPO is more likely if the VC firms participating in the financing are established investors. A more experienced VC firm is therefore more able to add value. For IPO exits, it may also make it easier to build a strong IPO syndicate. Below, we proxy experience by the age of the VC firm. As for the probability of liquidation, the outcome is less clear. On the one hand, one would expect better investment selection at the due diligence stage (making it more likely that bad investments are abandoned quickly); on the other hand, the better value adding along the development of the company reduces its likelihood.

Finally, to control for project-specific characteristics, we also include industry dummies for the most important industry sectors, the amount invested in the given round as well as dummies for stages of development. Moreover, in line with previous studies (Lerner, 1994; Gompers and Lerner, 1999; Jeng and Wells, 2000; Bottazzi and Da Rin, 2001; Cumming et al., forthcoming), we include measures of stock market liquidity in our estimations. This is particularly important as we also include observations from the Internet bubble. Next to its impact on IPOs and trade sales, this may also affect liquidations since venture capitalists could decide to more quickly redirect their effort into new projects (Kanniainen and Keuschnigg, 2003; Fulghieri and Sevilir, 2004).

⁶ The degree of innovation is also expected to be an important driver but unfortunately our data do not allow us to proxy the degree of innovation in a meaningful way.

⁷ We thank one of our referees for pointing this out.

3. Data

The data used in this paper has been extracted from the VentureXpert database (Thomson Financial). Our database is made up of successive records, each record pertaining to one investment round in a given venture-backed firm. The empirical analysis of the paper is done per round as we model the time-to-exit since a given financing round and taking into account the characteristics of the funding that took place at that time. The relevant variables are listed below. Observations cover the period from 1/1/1980 to 6/23/2003 (date at which the data has been collected). The data was pre-filtered to remove all records where the times-to-exit (DURATION variable thereafter) are smaller than 14 days or larger than 20 years and we also removed all records for which the amount of money received by the firm is smaller than \$10,000 or larger than \$100,000,000. These observations are meaningless outliers for which the recorded values do not belong to a plausible range. This pre-filtering leads us to discard very few records and gives us a sample made up of 22,042 investment rounds for 5817 distinct venture-backed companies.⁸ To characterize the firms in our dataset and the stage financing they received from venture capitalists, we use the following variables, split into ‘Entrepreneurial firm-related variables’ and ‘VC firm-related variables’ for ease of exposition.

3.1. Entrepreneurial firm-related variables

Industry type (dummy variable): INTERNET (internet industry), BIOTECH (biotech), COMPUTER (computer), SEMIC (semiconductor), MEDICAL (medical), COMMEDIA (communication and media) and OTHERIND (other industries than those listed above). These variables are equal to 1 if the given firm belongs to the specified industry.

Stage of development (dummy variable): EARLY (early stage), EXPANSION (expansion stage), LATER (later stage), BUYACQ (buy/acquisition stage), OTHERSTAGES (other stages than those listed above). Set to 1 if the financing stage matches the description of the variable.

Stage of technological development (dummy variable): TECHIMPROVEMENT. Set to 1 if the financing stage was characterized by a technological improvement, which we define as an advancement in the stage of development. For instance, a company that was in the early stage in the first round and in the expansion stage (or later stage) in the second round obtains a value of 1. This variable is only active for round 2 and above. We use this variable to proxy whether specific milestones have been achieved by the entrepreneurial firm.⁹

Type of exit (dummy variable): IPO (IPO exit), TRADESALE (trade sale exit), LIQUID (liquidation exit). Set to 1 if the firm exited according to the exit specified by the variable. Note that many firms are characterized by IPO = 0, TRADESALE = 0 and LIQUID = 0 as they are still ‘active’, i.e., venture capitalist have not yet exited, or

⁸ Note that we get the same qualitative results and conclude similarly when using non-filtered data. It is however well known that in some cases meaningless outliers can substantially affect the precision of the econometric estimations. We therefore focus on the filtered data.

⁹ We examined alternative variable specifications but their results were very similar to this TECHIMPROVEMENT variable. See Section 6.5 for a related discussion.

have exited via a fourth exit route.¹⁰ This will yield right-censored durations in the statistical analysis.¹¹

Geographical location of the entrepreneurial firm (dummy variable): WEST (California), NORTHEAST (Massachusetts, New York and Pennsylvania), SOUTH (Texas), and MIDWEST (Illinois and Ohio). Set to 1 if the location matches the description of the variable. This classification of regions is the same as in Lerner et al. (2005).

Total amount of money received by the firm at the given round (in millions of USD): AMOUNT.

Number of days elapsed between the date at which the round began and the exit date if there was an exit: DURATION. If the firm has not yet exited, this variable gives the number of days elapsed between the date at which the round began and the date of the data collection (June 23rd, 2003).¹² This variable is the main focus of our analysis as it characterizes the ‘life’ of the investment since a given round.

3.2. VC firm-related variables

Syndicate size, i.e., number of VC firms that participated in that financing round: SYNDSIZE.

Age (in years) of the oldest VC firm in the syndicate: AGEOLDESTVC.

Dummy variable equal to 1 if at least one VC firm in the syndicate is in the same US state as the entrepreneurial firm: SAMESTATEVC.

Dummy variable equal to 1 if at least one VC firm in the syndicate is in California: WESTVC.

Dummy variable equal to 1 if at least one VC firm in the syndicate is in the Northeast area of the US: NORTHEASTVC.

In addition, the selection variable (round number) is ROUND. This indicates which financing round we are dealing with. Finally, we also include a measure of stock market liquidity, namely the number of IPOs done in a given year on the most important US stock markets (NASDAQ, NYSE and AMEX). This measure is used in other studies on venture capital exits and was found to be an important variable for explaining exit choices. We denote this variable IPOACTIVITY. These data were taken from Jay Ritter’s web page.

4. Survival analysis and competing risks models

Our statistical analysis relies on survival analysis and competing risks models, which are powerful models tailored to model durations that end with multiple exits. They originate from the engineering sciences and have been extensively used in the medical sciences and in studies of labor markets. By definition, a competing risks model is also a generalization of

¹⁰ Because of the structure of the competing risks model used in the statistical analysis, the fact that we do not model explicitly these other types of exit routes does not lead to any bias in our estimations for the IPO, trade sale and liquidation exits.

¹¹ Note that if a company had more than one financing round, we consider the exit type at the very end of the financing cycle (i.e., the exit route chosen at the end of the last round). In other words, for a given firm, this variable takes the same value for all financing rounds.

¹² This is characterized as a right-censored duration in the terminology of survival analysis and does not lead to any estimation bias. See Section 4.

a binary exit model. In that latter class of models, one usually focuses on the occurrence of the exit (Logit model) or on the time until exit (with only one exit). In VC studies, such models have been put forth by e.g., Gompers (1995) and Cumming and MacIntosh (2001). It is important to stress that the competing risks approach is much richer, as it distinctively takes into account many possible type of exits. As such, each type of exit can feature its own dynamics and the covariates can influence the exit decisions separately. Duration models also take into account the fact that, at the time of the analysis, the exit may not have yet occurred. These yield right-censored durations. In contrast to more simple (but incorrect) OLS estimations, duration models thus explicitly take into account the information provided by the exits that have not yet occurred.¹³ In this section, we first detail a simple two-state competing risks model and then show how we use a multi-state competing risks model in the VC exits framework. Additional information on survival analysis and/or competing risks models can be found in Lee and Wang (2003).

4.1. A simple competing risks model

The next sub-section presents the competing risks model used in Section 6. We however first present a simple competing risks model with two exits (success and failure) to illustrate the general methodology. Let us consider a set of investments characterized by their durations (i.e., times until exit) and their exit types. For each project, we have a set of pairs (t, y) , where t is the duration of the investment and y is a dummy variable indicating the exit type. Let N_s denote the number of successful projects and N_f be the number of projects that failed ($N_s + N_f = N$, N being the number of projects in the dataset).¹⁴ The only two possible exits are characterized by mutually exclusive end states: $y = 1$ (success) or $y = 0$ (failure). For simplicity, let us assume that the hazard function is constant with respect to time. In survival analysis, the hazard function gives the conditional instantaneous probability of exit given that the subject at risk has not yet exited at that time.¹⁵

The idea of the competing risks model is to let the hazard rate vary with the end state. Thus we define λ_s (respectively λ_f) as the hazard rate of duration t when the project is a success (failure). In most competing risks models, the hazard rates are made dependent on a set of covariates. These can thus be viewed as explicative variables. For example, the exponential form $\lambda_s = e^{\beta_{s,0} + \beta_{s,1}X_1 + \dots + \beta_{s,k}X_k}$, where $X_1, \dots, X_k$ are the covariates, is widely used. The coefficients $\beta_{s,1}, \dots, \beta_{s,k}, \beta_{f,1}, \dots, \beta_{f,k}$ then allow an immediate assessment of the influence of the explicative variables on the exits. In this simplified example, the initial funding of the project could be a possible covariate: a positive $\beta_{s,1}$ would then indicate that a higher funding leads to an increased ‘success’ exit. At the end of the duration, either state $y = 1$ (success) or state $y = 0$ (failure) is realized. In the framework of a competing risks model, the duration corresponding to the state that is not realized is truncated. From a methodological point of view, this implies that the realized state will contribute to the likelihood function via its density function, while the truncated state contributes to the likelihood function via its survivor function. Because we assume a constant hazard in this

¹³ In other words, a project that has yet not exited does provide important information and it would be wrong to focus solely on the projects which have ended.

¹⁴ In the next subsection, we introduce a more complete model that also includes non-exited projects.

¹⁵ For example, in a medical science context, the hazard of death by heart attack at time t for a patient is the instantaneous probability of dying of a heart attack at time t given that the patient is still alive ‘just’ before time t .

example, the density function for successful exits is $\lambda_s e^{-\lambda_s t}$ and the survivor is $e^{-\lambda_s t}$; for failures, they are respectively $\lambda_f e^{-\lambda_f t}$ and $e^{-\lambda_f t}$. When estimating the parameters that drive successful exits (i.e., those involved in λ_s), we thus maximize the following likelihood:

$$L = \prod_{i=1}^{N_s} f_i \prod_{j=1}^{N_f} S_j, \quad (1)$$

where i indexes the successful projects, j indexes the projects that failed, $f_i = \lambda_s e^{-\lambda_s t_i}$ and $S_j = e^{-\lambda_s t_j}$. Similarly, $L = \prod_{j=1}^{N_f} f_j \prod_{i=1}^{N_s} S_i$, with $f_j = \lambda_f e^{-\lambda_f t_j}$ and $S_i = e^{-\lambda_f t_i}$, is maximized to estimate the parameters that enter λ_f . This discussion also shows that competing risks models focus on both the type of exits and time to exit (duration). In contrast, a Logit model for example would only focus on the type of exit (binary choice) and the likelihood function of a Logit model would not take into account the t variable. Further details regarding the construction of likelihood functions of competing risks models are available in Lee and Wang (2003).

4.2. A competing risks model for venture capital exits

The simplified example has already hinted at how firm-related variables can be linked to the exit process. Reframing the analysis in the VC framework, the competing risks model can be used to model the exit times and types of exit of VC-backed investments. We must first allow for multiple exits (IPO, trade sale and liquidation), which is an immediate generalization of the two-state model. Secondly, we must allow for the fact that some firms should be treated as non-exited firms. The model should accommodate the fact that we take a snapshot of the firms at the time of the data retrieval. Hence, some investments have indeed been exited while others are still in the pipeline. In our framework, the durations for these latter investments should be considered as being right-censored. Right-censoring is immediately taken care of by the model as an investment that has yet not exited is characterized by $IPO = 0$, $TRADESALE = 0$ and $LIQUID = 0$ and will only enter the likelihoods through its survivor function. Hence the model does not require any modification.

Let us illustrate how this is implemented by detailing the likelihood that is maximized. We concentrate on the IPO exit, but we proceed similarly for the other exits. For a given round, let N_{IPO} denote the number of firms that exited through an IPO and let i be the index which characterizes these firms; let j be the index which characterizes the other firms (there are $N - N_{IPO}$ such firms if we suppose that the total population of firms is N). By definition, when the likelihood is maximized for the IPO outcome, the durations for the firms (indexed by j here) which did not exit through an IPO are considered right censored and contribute to the likelihood through their survivor function. Hence the full likelihood is $L = \prod_{i=1}^{N_{IPO}} f_i \prod_{j=N_{IPO}+1}^N S_j$, where f and S are the density and survivor functions for the IPO outcome respectively (the density function used in the paper is detailed below).

It should be stressed that we estimate the competing risks models for a given round. This stems from the fact that a duration is defined herein as the time elapsed between the actual round date when the firm received the funding from the venture capitalist and the exit date (if the firm has exited). If the firm has not yet exited, the duration is right-censored and the duration is defined as the time between the round date and the date of the data collection (June 23rd, 2003). Regarding the choice of the density function,

non-monotonically increasing or decreasing hazards should be endorsed, i.e., we should use a density distribution such as the generalized Gamma density distribution. More specifically, this density distribution is shaped by three parameters κ , σ and μ , and is specified as

$$f(t, z, \sigma) = \frac{\gamma^\gamma}{\sigma t \sqrt{\gamma} \Gamma(\gamma)} e^{(z\sqrt{\gamma} - u)} \quad (2)$$

if $\kappa \neq 0$, and

$$f(t, z, \sigma) = \frac{1}{\sigma t \sqrt{2\pi}} e^{(-z^2/2)} \quad (3)$$

if $\kappa = 0$, where $\gamma = |\kappa|^{-2}$, $z = \text{sign}(\kappa)(\ln(t) - \mu)/\sigma$, $u = \gamma e^{(|\kappa|z)}$. The dependence with respect to the covariates is introduced through $\mu_k = X_k \beta$, where k is the observations' index. The κ and σ parameters affect the shape of the hazard function. More precisely, it can be shown that the generalized Gamma density distribution encompasses many density distributions. For example, if $\kappa = 1$ and $\sigma = 1$ it simplifies into the exponential distribution; if $\kappa = 1$, then it is the Weibull distribution. Hence it is the combination of κ and σ that allows flexibility. From an economic point of view, a constant hazard (exponential distribution) implies that the time elapsed does not bear on the exit probabilities (e.g., an exit is as likely after 1 month as after 1 year). A Weibull model yields monotonically increasing or decreasing hazard. Monotonically increasing or decreasing hazard functions imply that, as time flows, the likelihood of exiting gets either larger and larger or smaller and smaller. In our context, the possibility of hump-shaped hazard rates call for a very flexible density distribution, which justifies our choice of the generalized Gamma density distribution.

Finally, let us explain how the explicative variables enter the density function. In the VC framework of this study (IPO, trade sale and liquidation exits) and if all explicative variables detailed in Section 3 are included in the model, this translates into three specifications for μ_k as we have three mutually exclusive exit possibilities¹⁶

$$\mu_{k,\text{IPO}} = \beta_{1,\text{IPO}} \text{INTERNET}_k + \beta_{2,\text{IPO}} \text{BIOTECH}_k + \dots, \quad (4)$$

$$\mu_{k,\text{TS}} = \beta_{1,\text{TS}} \text{INTERNET}_k + \beta_{2,\text{TS}} \text{BIOTECH}_k + \dots, \quad (5)$$

and

$$\mu_{k,\text{LIQ}} = \beta_{1,\text{LIQ}} \text{INTERNET}_k + \beta_{2,\text{LIQ}} \text{BIOTECH}_k + \dots \quad (6)$$

A significantly *negative* value for any β parameter implies that an *increase* in the corresponding variable leads to a significantly *faster* exit. For example, a significantly negative coefficient of the SYNDSIZE variable in the specification of the IPO exit would mean that, as the size of the syndicate grows, the time-to-exit for an IPO gets shorter. Note that, for

¹⁶ Note that in practice more than three exit types are observed. This is also the case in our sample. We focus however on the IPO, trade sale and liquidation exits as these are the most important exits and are the real focus of our analysis. Because of the multiplicative nature of the likelihood function for competing risks models (see Eq. (1)), not modelling explicitly the other (few) types of exits does not lead to any bias.

dummy variables, exponentiated coefficients (i.e., $e^{\beta_{1,IPO}}$ for example) have an easy interpretation as time ratios. These time ratios can then be directly compared with each other, yielding relative time ratios. The latter are very easy to interpret as a relative time ratio indicates how fast/slowly the change of category impacts the (conditional) exit probability. For example, the relative time ratio (for the IPO exit) of the internet industry with respect to the biotech industry is equal to $e^{\beta_{1,IPO}}/e^{\beta_{2,IPO}}$. We further provide examples when we discuss the empirical results.

5. Descriptive analysis

In this section, we provide a descriptive analysis of our dataset. Estimation results for the competing risks model are given in the next section. The main summary statistics for our dataset are presented in Table 1. Table 2 provides the frequency of exit routes for different types of investment stages and Table 3 gives a breakdown of key statistics by investment rounds, industries and stages of development.

From Table 1 we see that, for internet investments, the fraction of deals that exited through a liquidation is quite large compared to its relative importance (29.8% of all liquidations are internet investments while exited internet deals only account for 11.5% of the total sample). It may be the case that such investments are stopped more quickly than in other industries in order to redirect human resources within VC funds towards more promising investments (Fulghieri and Sevilir, 2004). This may also be due to the fact that many young internet firms missed the internet bubble period and therefore could no longer successfully exit due to a loss of investors' interest. Second, while firms that had an IPO experienced more technological improvements in follow-up rounds (TECHIMPROVEMENT variable) than those that had a trade sale (32.6% versus 29.6%), the duration of investment (DURATION variable) was also shorter for IPOs than for trade sales (a breakdown of duration along several dimensions is provided in Table 3). As mentioned earlier, this suggests that IPO candidates may on average be more profitable than candidates for a trade sale. Moreover, VC-backed firms that eventually were liquidated exhibited a technological improvement in 33.5% of the follow-up rounds, i.e., more than for IPOs and trade sales. While it may be surprising at first sight, this is mainly due to the fact that we do not observe final rounds for which we expect liquidated firms to almost never realize an improvement.

Third, it is instructive to compare exited with not-yet-exited investment rounds with respect to geographical location of VC-backed firms. Those located in the West and Northeast regions have a slightly higher proportion of exited rounds (47.4% and 18.9% versus 40.1% and 16.2%) than in any other region. This is in line with our prediction that VC-backed firms located in developed entrepreneurial clusters such as Silicon Valley and Route 128 obtain better access to opportunities and needed resources. A similar pattern is observed for the location of VC firms (WESTVC and NORTHEASTVC variables). Overall, about half of the entrepreneurial firms in our sample are located in California (WEST variable). The next largest region is Northeast. About one third of the investments in our sample are from other regions, which also include all those outside the USA. This contrasts with the location of VC funds that are much more concentrated in the West and Northeast regions. Indeed, these latter two regions account for almost 80% of the VC funds located in the USA (Lerner et al., 2005). There is little variation in the different subsamples in terms of VC firm characteristics. The only meaningful differences arise

Table 1

Dataset: main summary statistics (mean of variable)

Variable	First round	All rounds (full sample)	Exited rounds only					Unexited rounds only
			All routes	IPO	Trade sale	Liquidation	Others	
<i>Industry dummies</i>								
Internet	0.169	0.146	0.115	0.113	0.094	0.298	0.035	0.186
Biotech	0.050	0.065	0.085	0.134	0.068	0.021	0.077	0.040
Computer	0.265	0.297	0.316	0.278	0.349	0.193	0.403	0.274
Medical	0.106	0.122	0.115	0.120	0.112	0.082	0.179	0.132
Semic	0.078	0.084	0.098	0.125	0.092	0.080	0.031	0.067
Commedia	0.130	0.144	0.163	0.141	0.177	0.208	0.058	0.120
Otherind	0.201	0.140	0.108	0.089	0.108	0.118	0.218	0.181
<i>Stage of development</i>								
Early stage	0.646	0.338	0.340	0.362	0.333	0.331	0.304	0.336
Expansion stage	0.180	0.392	0.372	0.381	0.365	0.399	0.355	0.418
Later stage	0.043	0.199	0.232	0.210	0.241	0.220	0.387	0.156
Buyout/acquisition	0.097	0.043	0.035	0.028	0.037	0.036	0.044	0.054
Other stages	0.022	0.017	0.011	0.012	0.013	0.001	0.002	0.026
TECHIMPROVEMENT	–	0.311	0.306	0.326	0.296	0.335	0.246	0.318
DURATION (days)	–	–	1516	1241	1675	1199	2041	–
<i>Geographical location of VC-backed firm</i>								
West	0.394	0.442	0.474	0.466	0.482	0.475	0.424	0.401
Northeast	0.176	0.177	0.189	0.184	0.200	0.156	0.160	0.162
South	0.044	0.044	0.038	0.045	0.029	0.052	0.087	0.052
Midwest	0.028	0.023	0.018	0.018	0.018	0.025	0.010	0.031
Others	0.357	0.313	0.281	0.287	0.272	0.292	0.320	0.355
<i>VC firm characteristics</i>								
SYNDSIZE (# investors)	2.937	3.964	4.242	4.483	4.123	4.279	4.012	3.602
AGEOLDESTVC (years)	25.755	29.854	28.857	28.899	27.656	35.978	29.357	31.154
SAMESTATEVC	0.499	0.659	0.695	0.679	0.700	0.708	0.709	0.612
WESTVC	0.489	0.643	0.682	0.668	0.685	0.691	0.723	0.593
NORTHEASTVC	0.482	0.614	0.645	0.652	0.645	0.635	0.620	0.575

This table provides averages of variables for various sub-samples: first round investments only, the full sample (i.e., all rounds), exited rounds only (including each exit route separately), and un-exited rounds only.

Table 2
Frequency of exit route for different types of investment stage

Stage of investment	No. obs.	Exit route				Ratio TS-IPO
		IPO (%)	Trade sale (%)	Liquidation (%)	Other routes (%)	
<i>Panel A: First investment round (Round = 1)</i>						
Early stage	1839	33.8	53.0	9.8	3.4	1.57
Expansion stage	472	38.4	50.4	8.7	2.5	1.31
Later stage	141	34.8	55.3	5.0	5.0	1.59
Buyout/acquisition	218	28.4	56.0	8.3	7.3	1.97
Other stages	54	31.5	64.8	1.9	1.9	2.06
<i>Panel B: All investment rounds</i>						
Early stage	3957	35.1	52.4	8.4	4.0	1.49
Expansion stage	4397	33.6	52.8	9.1	4.5	1.57
Later stage	2692	30.0	56.2	8.3	5.5	1.87
Buyout/acquisition	407	26.3	57.7	10.1	5.9	2.20
Other stages	249	30.1	62.7	5.2	2.0	2.08

Panel A gives the exit routes frequencies by stage of investment for the first investment round (by focusing on the first round only, we make sure that each exited company is represented once). Panel B provides similar summary statistics for all investment rounds of exited companies. Column 2 gives the number of observations per stage of investment for which an exit already occurred. The last column gives the ratio of trade sales over IPOs. Since we exclude yet-to-exit investments, the total number of observations is 2724 for Panel A and 11,702 for Panel B.

between the first round subsample and the full sample. But this is not surprising, given the way our variables are defined and that syndicate size tends to increase in subsequent rounds.

Table 2 shows that the proportion of exit types is quite similar across financing stages, except from the fact that there is a slight increase in trade sales with the increasing stage of development (and the decreasing likelihood of IPOs). In both panels, the ratio of trade sales over IPOs therefore tends to increase. Furthermore, this ratio is always greater than 1. For instance, there are about 50% more trade sales than IPOs for early stage investments.

A breakdown of AMOUNT and SYNDSIZE by round number (Panel A of Table 3, shown for up to round 5) shows that the AMOUNT variable increases steadily when going from round 1 to round 4. From round 3 onwards, it however stabilizes around \$9 million. The fact that firms receive a much lower amount of money in their first round of financing is consistent with the literature: venture capitalists do not want to commit too many funds at the start of the venture capital process. The SYNDSIZE variable also seems to be lower for the first rounds. For all types of exits, the duration decreases as the number of rounds increases, which is to be expected and is in line with the literature that conjectures a reduction in duration as the project is developed. For example (IPO exit), the mean duration goes from 1620 days to 925 days as the representative firm goes from round 1 to round 5 (for trade sales, it decreases from 2059 days to 1506 days; for liquidations, it decreases to 981 days from 1554 days). Again, there is a variability in the means for all the exit routes.

The breakdown of firms across industries (Panel B of Table 3) shows that internet and computer companies attract the substantial part of the venture capital invested, while bio-tech companies are much less represented (in absolute number). Similarly, a breakdown of AMOUNT and SYNDSIZE by type of industry (Panel B of Table 3) shows that the

Table 3

Summary statistics for the investment rounds, industries and stages of development

Variable	No. obs.	Amount	Syndsize	Duration (in days)			
				IPO	TS	IPO and TS	Liquidation
<i>Panel A: Breakdown by investment rounds</i>							
All rounds	22,042	7.7 (12.0)	3.9 (3.3)	1219 (1066)	1666 (1335)	1496 (1259)	1203 (1035)
1st round	5817	6.5 (10.7)	2.9 (2.1)	1620 (1148)	2059 (1447)	1887 (1355)	1554 (1168)
2nd round	4691	8.1 (11.7)	3.8 (3.0)	1350 (1093)	1739 (1355)	1586 (1273)	1286 (1049)
3rd round	3562	9.4 (13.0)	4.5 (3.6)	1125 (995)	1603 (1331)	1414 (1232)	1087 (960)
4th round	2548	9.7 (14.1)	4.7 (4.0)	956 (977)	1514 (1277)	1290 (1197)	996 (852)
5th round	1765	8.6 (13.2)	4.6 (4.0)	925 (954)	1506 (1294)	1286 (1210)	981 (845)
<i>Panel B: Breakdown by industries</i>							
INTERNET	3502	12.9 (15.5)	3.9 (3.1)	670 (566)	991 (826)	852 (742)	721 (519)
BIOTECH	1468	6.9 (10.0)	4.1 (3.5)	1097 (860)	2062 (1370)	1523 (1212)	1354 (868)
COMPUTER	6352	5.9 (9.0)	4.0 (3.4)	1251 (1,071)	1599 (1325)	1486 (1259)	1254 (997)
SEMIC	1793	7.2 (11.4)	4.3 (4.0)	1725 (1376)	1835 (1387)	1787 (1383)	1562 (977)
MEDICAL	2673	5.5 (7.9)	3.9 (3.2)	1162 (890)	1792 (1249)	1542 (1162)	1684 (1407)
COMMEDIA	3169	9.5 (14.5)	4.3 (3.6)	1206 (1029)	1683 (1438)	1526 (1337)	1291 (901)
OTHERIND	3085	6.3 (12.0)	2.9 (2.7)	1504 (1274)	1943 (1351)	1794 (1341)	1624 (1525)
<i>Panel C: Breakdown by stage of development</i>							
Early stage	7427	5.1 (7.7)	3.5 (2.8)	1581 (1073)	1907 (1377)	1776 (1274)	1470 (1000)
Expansion stage	8827	9.3 (12.9)	4.2 (3.5)	1106 (1038)	1530 (1294)	1366 (1219)	1000 (932)
Later stage	4273	8.1 (13.3)	4.3 (3.8)	776 (840)	1400 (1161)	1183 (1101)	985 (835)
Buyout/acquisition	936	13.7 (18.0)	3.0 (2.3)	1237 (1134)	1790 (1268)	1617 (1253)	2127 (1763)
Other stages	579	3.8 (10.9)	2.5 (2.6)	1464 (1281)	2870 (1796)	2414 (1771)	1453 (1761)

Key statistics for different investment rounds, industries and stages of development. AMOUNT is expressed in \$1,000,000s and gives the amount of money received by the firm. SYNDSIZE is the size of the syndicate. Standard deviations are reported below each value in brackets.

average internet firm was given much more money (around \$12.9 million) than the other types of firms. Communication/media firms rank second (mean of \$9.5 million), while the other firms received on average around \$6–9 millions. The mean of SYNDSIZE does not really change across industry types. Focusing on IPO exits only, it is obvious that internet firms had the fastest exit, with a mean of 670 days. Firms in the other industries

needed much more time, the slowest being the semiconductor firms (mean duration of 1725 days). When focusing on liquidations, it is also true that internet firms had the fastest exits (the representative internet firm exhibits a mean duration to liquidation of around 721 days).

Regarding the pattern of AMOUNT and SYNDSIZE for the different financing stages (Panel C of Table 3), buyouts/acquisitions provide the largest mean amount (around \$14 million) and involve on average three venture capital firms. In contrast, early stage investments are characterized by an average amount of \$5.1 million and an average of 3.5 venture capital firms. For IPO exits, a breakdown of DURATION per financing stage yields a mean of 1581 days (early stage), and decreases as we go to the expansion and later stages. Furthermore, we observe similar patterns for trade sale and liquidation.

6. Estimation results

As indicated in Section 4, we use the generalized Gamma density function as the distribution for the underlying error term.¹⁷ We also allow for heterogeneity in our model.¹⁸ When dealing with data such as ours which span many firms distributed in a collection of industries, we suspect that it features some unobserved heterogeneity. Hence, although our database features many important firm-specific and VC-specific variables, we should take into account the possibility that some piece of information may be left out. From an econometric point of view, this could lead to heteroscedasticity in the residuals and should be corrected. Actually this is quite close to using the well-known White's heteroscedastic-consistent estimators in a time-series framework.¹⁹

To check the robustness of our results, we also estimated a proportional hazard model ('Cox model') with the same explanatory variables. Results were qualitatively similar. Finally, we also estimated a binary model where the duration to exit (either IPO, or trade sale, or liquidation) is modelled. As explained in Section 4, this model is constrained with respect to the competing risks model and brings forth much less information. Nevertheless it delivered similar insights into the role of most explanatory variables, although the richness of the full model was lost.

As far as the competing risks model is concerned, Table 4 displays the estimation results for the first, second and third rounds. This table features two panels: the top panel presents the estimation results for the entrepreneurial firm-related variables, while the bottom panel displays the outputs for the VC firm-related variables.

¹⁷ We avoid multicollinearity problems by not including the constant and the OTHERSTAGES dummy variable.

¹⁸ Allowing for heterogeneity leads to somewhat less efficient estimators when dealing with small datasets. We however have a very large dataset and the minor loss of efficiency is irrelevant here. Note that we conclude similarly (from a qualitative point of view) with and without the frailty estimation option.

¹⁹ In this paper, we allow for heterogeneity by estimating the model with the 'frailty' option. The residuals of the models are analyzed after each estimation. As suggested by the literature on survival analysis, we focus on the generalized Cox–Snell residuals: if the model fits the data well, then these residuals should be exponentially distributed. This can be checked by plotting their cumulative hazard function, along with the benchmark line with slope equal to 1. Such residual analysis also indicated that it was sensible to allow for heterogeneity.

Table 4
Estimation results for the competing risks model

Coefficient	First round			Second round			Third round		
	IPO	Trade sale	Liquidation	IPO	Trade sale	Liquidation	IPO	Trade sale	Liquidation
<i>Entrepreneurial firm-related variables</i>									
INTERNET	9.163***	9.144***	9.611***	8.412***	8.277***	9.522***	9.170***	8.618***	10.224***
BIOTECH	8.927***	9.457***	11.121***	8.213***	8.662***	10.646***	8.601***	9.020***	12.039***
COMPUTER	9.299***	9.470***	10.956***	8.627***	8.638***	10.644***	9.344***	9.130***	11.397***
SEMIC	9.254***	9.120***	10.700***	8.683***	8.277***	10.577***	9.582***	8.610***	11.576***
MEDICAL	9.135***	9.262***	10.693***	8.464***	8.381***	10.479***	9.136***	8.843***	11.009***
COMMEDIA	9.264***	9.027***	10.197***	8.428***	8.136***	9.956***	9.372***	8.561***	10.895***
OTHERIND	9.790***	9.643***	10.999***	9.060***	8.855***	10.431***	9.941***	9.225***	11.552***
EARLY	−0.547*	−0.326*	−1.050**	0.135	0.133	−0.606	0.039	−0.448*	−0.679
EXPANSION	−0.776**	−0.304*	−0.989*	−0.076	−0.025	−0.880	−0.077	−0.462*	−0.877
LATER	−1.701***	−0.633***	−0.997*	−0.499	−0.194	−0.996	−0.321	−0.581**	−1.005
BUYACQ	−0.341	−0.199	−0.521	−0.002	−0.107	−0.316	0.378	−0.844***	−0.921
TECHIMPROVEMENT				−0.174**	−0.161***	−0.267**	−0.150	0.098	0.142
AMOUNT	−0.011***	−0.004	−0.017***	0.010**	−0.001	−0.023***	−0.008*	−0.004	−0.037***
WEST	−0.026	−0.060	−0.017	−0.014	0.095	−0.021	−0.092	0.080	0.109
NORTHEAST	−0.014	−0.095	−0.142	0.105	−0.039	−0.235	0.004	0.036	0.018
SOUTH	0.040	0.122	−0.226	−0.212	0.440***	−0.246	0.052	0.590***	−0.015
MIDWEST	0.371	0.070	−0.379	0.575**	0.249	−0.723	0.335	0.386*	−0.261
<i>VC-related variables</i>									
SYNDSIZE	−0.062***	−0.024**	−0.054*	−0.031**	0.002	0.014	0.002	0.023**	0.106***
AGEOLDESTVC	0.004	0	−0.016***	0.001	−0.003	−0.019**	−0.008	0.001	−0.015
SAMESTATEVC	0.084	−0.114**	−0.064	0.104	−0.170***	−0.085	−0.009	−0.155*	−0.024
WESTVC	−0.001	−0.050	0.064	0.033	−0.151**	−0.155	−0.158	−0.178**	−0.405**
NORTHEASTVC	−0.001	−0.098**	0.236**	−0.157*	−0.111**	0.100	−0.177*	−0.123*	−0.261
IPOACTIVITY	−0.001***	−0.001***	−0.001***	−0.001***	−0.001***	0	−0.002***	−0.001***	0

Estimated coefficients for the competing risks model with three exits (IPO, trade sale, liquidation). We model the times-to-exit (durations) which start at the given financing round and end when there is an exit. Non-exited firms are taken into account as the model treats their durations as being right-censored at the date of the analysis. The underlying density distribution for the durations is the generalized Gamma density distribution and we allow for heterogeneity (frailty). A ***, ** and * indicates that the coefficient is significant at 1%, 5% and 10% respectively.

6.1. IPO exit

Results in the top panel of Table 4 show that later stage investments exit more quickly than expansion stage investments. This is also the case for expansion and later stage investments with respect to early stage investments, which is in accordance with the literature review.²⁰ For all three rounds, larger committed amounts also decrease exit times (significantly negative AMOUNT coefficient).²¹ For example, simulations for the median exit time show that increasing the funding in round 1 from \$10 million to \$50 million leads to a 37% decrease in the median exit time.

Biotech firms have the fastest exits and are followed by internet and medical firms. To illustrate with the first round, with respect to biotech firms, computer firms exhibit a relative time ratio of almost 1.45 (computed as $e^{9.299}/e^{8.927}$) while the other industries are in-between (not taking into account the ‘other industries’ category). More generally, estimation results for the industry variables indicate that there is a strong industry effect. As far as the geographical location of the entrepreneurial firm is concerned, there are no significant differences between the US regions, although firms in the Midwest area do exit much less frequently.

Regarding VC firm-related variables, larger syndicate sizes increase the hazard for IPOs, and thus decrease exit times, as the SYNDSIZE coefficient is significantly negative (round 1 and 2). Simulations run for the median exit time as a function of syndicate size are informative: increasing the syndicate size from 2 to 4, and then from 4 to 8, results in median exit time decreases of about 13% and 21% respectively. This supports the hypothesis that more syndicate partners help add value. In terms of IPOs, this may include attracting well-established underwriters into the IPO syndicate. Further variables related to value-adding such as VC firm experience (AGEOLDESTVC) and geographical proximity (SAMESTATEVC) do not impact the likelihood of an IPO, nor does the VC firm’s location (WESTVC and NORTHEASTVC).

General market conditions have a significant impact on the IPO probability (as well as on all other exit routes). Indeed, the IPO volume (IPOACTIVITY) in a given year significantly reduces the time for an IPO. To illustrate, simulations indicate that switching from a ‘quiet’ IPO market to a very active IPO market can decrease the median exit time (since the first round of financing) of a typical firm by about 30% (we look at the effect of increasing the IPO activity by a factor of five). This is in line with earlier studies on the liquidity of stock markets and supports the idea that stock markets are an important ingredients for venture capital markets as it makes exits easier (which reduces risk).

The model provides important information on the dynamics of the exit process. More specifically, the model rejects the null hypothesis of monotonically increasing or decreasing hazard. This is shown in the top left panel of Fig. 1, where we plot the estimated hazard function for the first four industry classifications for a typical venture capital-backed firm that would receive (at the early stage) a \$10 million funding provided by a syndicate of four venture capitalists. Regarding the shape of the hazard functions, one has first a

²⁰ For the first round, this is confirmed by Wald statistical tests, according to which the null hypotheses $\text{coef(EARLY)} > \text{coef(EXPANSION)}$ and $\text{coef(EXPANSION)} > \text{coef(LATER)}$ are not rejected individually.

²¹ We also considered scaling the variable AMOUNT to accommodate for industry differences. We used the total amount of money received by a VC-backed firm at a given round (AMOUNT) in excess of the average amount of money received by VC-backed firms in that industry. We then obtained similar results.

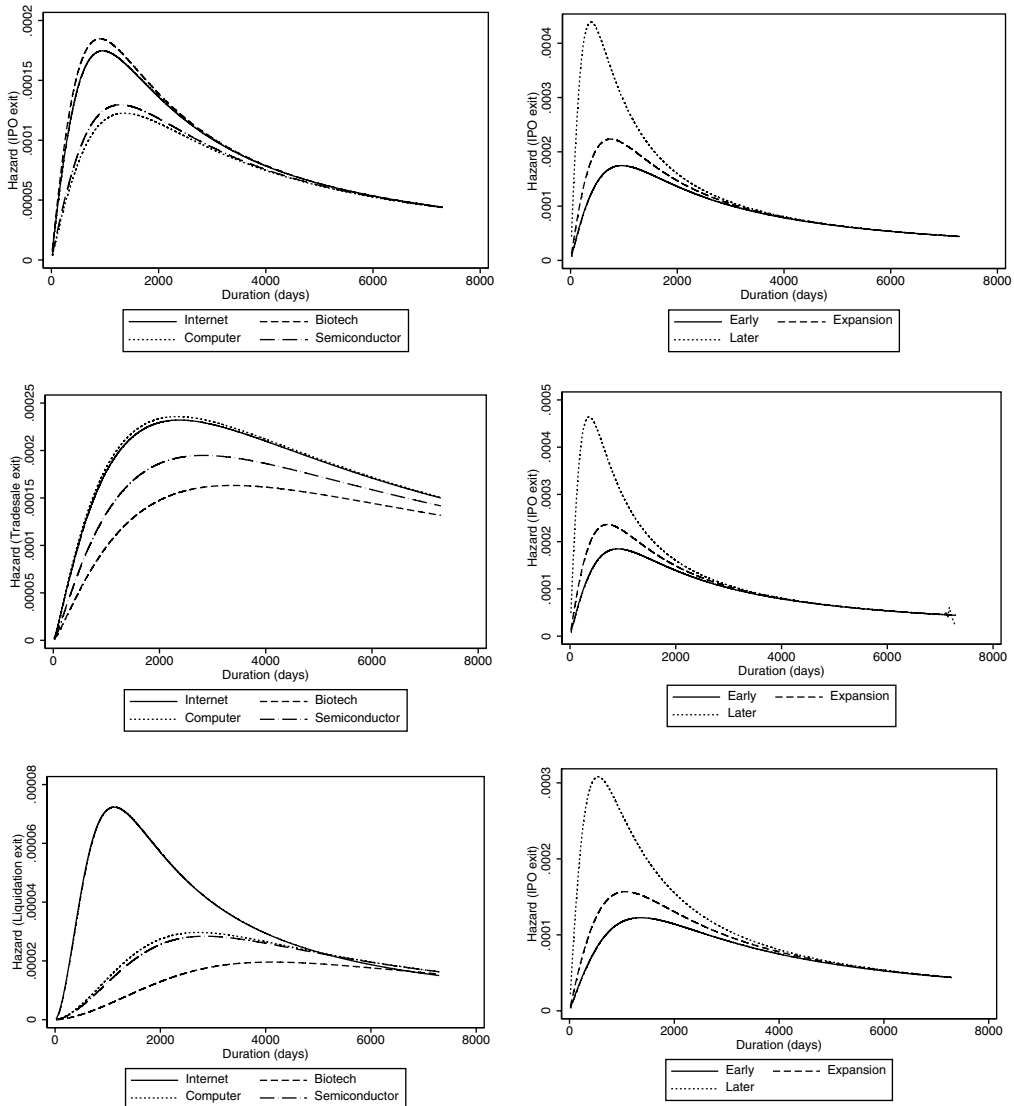


Fig. 1. Left panel, top to bottom: hazard functions for the IPO, trade sale and liquidation exits as a function of industry type. Right panel, top to bottom: hazard functions (IPO exit) for the internet, biotech and computer industries as a function of the type of stage. All durations start at round 1. For all graphs, the x -axis denotes the number of days elapsed (since round 1), while the y -axis gives the conditional exit probability (i.e., hazard rate) via the specified route.

sharply increasing hazard (to about 1000–1500 days) and then a slowly decreasing hazard. Thus, as time flows, VC-backed firms first exhibit an increased likelihood of exiting to an IPO. However, after having reached a plateau (around 1000–1500 days of existence, i.e., 2.75–4.0 years), investments that have not yet exited have fewer and fewer possibilities of exits as time increases. This pattern is stronger for biotech and internet firms which tend

to reach their plateau sooner than computer or semiconductor firms (around 5 years (1800 days) for these latter firms, around 3.3 years (1200 days) for the former). In the top right panel of Fig. 1 we plot the hazard functions for an internet firm and the three financing stages. As expected, the maximum of the hazard functions shifts left as we go from early to expansion and finally later stage financing. We then repeat the exercise for a biotech and computer firm, and the results are given in the middle and bottom right panels of Fig. 1.

6.2. Trade sale exit

We earlier hypothesized that candidates for a trade sale are less homogeneous than those for an IPO. While we do not provide a formal test, our results are supportive of this idea. The relative time ratios between the different industries are not as dispersed and belong to a tighter range. The classification is also different as the internet, semiconductor and communication/media firms have the fastest exit to a trade sale. The plots of hazard functions in the middle left panel of Fig. 1 tell the same story (same covariates as before). Note that in this case all hazard functions reach their maximum much later (around 2500–4000 days, i.e., 6.8–11 years) and decrease much more slowly thereafter. A comparison of hazard functions for exits to IPO and trade sale suggests that VC-backed firms first aim for an IPO exit and then consider (or are forced to consider) trade sale exits as their second choice. The multivariate results on stages of development further supports this conjecture. While the coefficients are smaller (in absolute terms) than for an IPO in the first two rounds, they suddenly become stronger (and significant) in the third round. In contrast to IPO exits, the effect of the AMOUNT variable is not significant. However, the effect of TECHIMPROVEMENT is similar as well as the IPO volume of the US market (IPOACTIVITY).

Finally, the geographical location of the entrepreneurial firm affects the trade sale dynamics. Overall, trade sale exits seem to be different for, on the one hand, West and Northeast firms and, on the other hand, South and Midwest firms. Indeed, firms face easier trade sale exits in the West and Northeast regions than for the other two regions, especially in follow-up rounds (the coefficients of SOUTH and MIDWEST become (weakly) significantly positive). This is reinforced by the local presence of VC firms in the same region that accelerate trade sales especially in West and Northeast.²²

6.3. Liquidation exit

The relative time ratio of internet firms with respect to the other firms is striking as it is between 1/3 and 1/4! This is clearly shown in the bottom left panel of Fig. 1 (same covariates as before) as there is a clear gap between the hazard functions of internet related firms and the other types of firms. Note also that the hazard function for internet firms quickly reaches its plateau (around 1200 days) and strongly decreases thereafter. In contrast, biotech firms are the slowest to liquidate. Lerner (1994) notes that “biotechnology firms [...] mature slowly and do not incur large up-front costs in building manufacturing facilities”,

²² Simulations for the median exit time tell the same story. For example (round 2), the median exit time decreases by about 13% when we compare a firm located in the West or Northeast regions with a firm located elsewhere. Similarly, the local presence of the VC firm yields median exit time decreases of about 15% (for VC-backed firms located in the West or Northeast regions).

which could explain why (in conjunction with the often lengthy Food and Drug Administration (FDA) approval process) these firms do not tend to liquidate quickly.

Technological progress (TECHIMPROVEMENT) also makes liquidations more likely. In fact, the effect seems to be strongest for liquidations than for IPOs or trade sales. Moreover a larger syndicate size accelerates liquidations. This result is rather surprising. A possible explanation is that the way our proxy is defined mainly relates to technological aspects of product development than business aspects (that focus on profitability). Indeed, achieving a technological breakthrough may also give information on how the product will look and thus whether it will ultimately be profitable. This dual view (business risk versus technological risk) is nicely captured by the theoretical framework of Berk et al. (2004). Another possibility that we cannot exclude a priori is that our proxy does not capture well the right technological progress. Indeed, TECHIMPROVEMENT indicates whether a milestone has been achieved in the previous round and not in the round we are examining. We therefore implicitly assume that having achieved important milestones in earlier rounds makes the progress in follow-up rounds also more likely. Possibly this implicit assumption does not hold. Unfortunately, it is impossible to create better proxies on milestones. We therefore need to be careful in interpreting our results on technological achievements.

Regarding the geographical location of the entrepreneurial firm, firms in the South and especially Midwest regions tend to liquidate much faster than firms in the West and Northeast.²³ The difference is sharply significant, as shown by the time ratios which are almost halved when switching from a West firm to a Midwest firm (e.g., at round 2, the time ratio would be $e^{-0.723}/e^{-0.021} = 0.496$).

6.4. A brief discussion of related results

The analysis based on competing hazard models has provided a number of interesting results regarding the dynamics of the exit pattern for VC-backed firms. Some of these results are strongly related to previously reported findings in the literature on venture capital and exit strategies by venture capital-backed firms. We briefly discuss these here and stress the contribution of our analysis to this recent strand of the literature.

Gompers (1995) provides evidence that the degree of asymmetric information has a significant impact on the time-to-exit. This implies that early stage investments require a longer lasting involvement of VC funds than later-stage investments, since asymmetric information decreases along with the reduction of technological risk. Our paper provides further evidence in line with this rationale. Cumming and MacIntosh (2001) find that exits occur more quickly for early stages of development, which they interpret as the result of a selection process to sort out the bad from the good projects. Our study provides a somewhat different picture. While exits from later-stage investment are quicker than for early stage investments (irrespective of the type of exit route) at the time of deal initiation, the results are mixed for later-round investments. Finally, Das et al. (2003) look at cumulative probabilities of exits. Among other things, they find that the likelihood of a trade sale increases with the stage of development (“this may be because many early staged firms that were unable to make it to the IPO stage settled instead for a buyout”). They further

²³ The coefficients of SOUTH and MIDWEST significantly differ from those of WEST and NORTHEAST.

show that successful companies in biotech and medical sectors exit more frequently. Our study goes a step further and looks at the time dimension of the exit process, how the exit probabilities evolve over time and how the dynamics of the exit process is affected by the actual outcome (IPO, trade sale or liquidation). In particular, the most striking feature is the difference between trade sales and IPOs (the main exit routes). While probabilities do not change that much in the first case, the probability of doing an IPO exhibits a strong, inverse U-shaped pattern; it increases very quickly and again decreases sharply right after it peaked. Successful companies that could not go public sufficiently quickly have to rely on other exit routes like trade sales.

In line with related work, the liquidity of exit markets, especially stock markets, are an important macro-economic factor affecting exit outcomes. This supports the hypothesis of Black and Gilson (1998), who argue that active stock markets allow venture capitalists to exit more easily because an IPO allows entrepreneurs to regain control of their company. As such, it provides an important exit channel for successful investments. This further supports the conjecture that more liquid stock markets accelerate exits and facilitate business creations through better recycling of informed capital, as argued by Michelacci and Suarez (2004). Our results are also in line with studies by Gompers et al. (2005) and Cumming et al. (forthcoming) that evidence how VC fund managers adapt their investment behavior to changes in stock market activities. Our paper identifies how IPO volume affects the dynamics of exit choices by accelerating time-to-exit for all exit routes, but especially for IPOs and trade sales. We also used a dummy variables equal to one if the investment was made during the IT bubble period of 1998–2000 instead of IPO volume. Results were qualitatively similar.

6.5. Robustness, limitations and extensions

We performed a number of robustness checks (not provided in the paper). For instance, we use alternative measures of VC firm experience such as the highest fund sequence number of the VC firms involved in the financing. Results were not materially different from what obtained with AGEOLDESTVC. We also used several alternative definitions of technological progress. For instance, we create an ordinal variable that captures the change in the number of stages of development from one round to another. As such, TECHIMPROVEMENT is only a binary transformation of this alternative variable. However, all our alternative measures were strongly correlated with the one we use in this study. We therefore only report TECHIMPROVEMENT as it is the one that, to our view, is the most intuitive one. Another possibility is that the presence of non-independent funds such as corporate funds and bank-affiliated funds could affect the quality of value-adding the syndicate can provide to entrepreneurial firms. However, we did not find significant effects in our sample. We also constructed a measure of syndicate heterogeneity but again this was not significant. Finally, as mentioned at the start of this section, we also estimated alternative multivariate models (Cox models), which yielded similar qualitative results, and a constrained binary outcome (exit/no exit) model.

7. Conclusion and outlook

Our results shed light on the competing exit possibilities for venture capitalists and on the dynamics of the time-to-exit for the IPO, trade sale and liquidation exits. For venture

capitalists, the decision to exit includes two main dimensions, the type and the timing of the exit. This paper has examined both dimensions of exit simultaneously in the framework of competing risks models. Besides the rigorous statistical modelling of exits times, this approach allows the computation of the instantaneous probabilities (hazard rates) of the different exit routes, conditional on the time already elapsed and on covariates included in the model.

Our empirical analysis delivers a series of interesting results. First, the hazard rates for IPO exits are clearly non-monotonic. While hazard rates for trade sale exits are also hump-shaped, our analysis suggests an exit order (IPO, and then possibly a trade sale) that is consistent with the fact that venture capitalists first target the IPO as the preferred way of cashing out on investments. Because the window of opportunity for trade sales extends for a considerable amount of time, trade sale exits are second-best choices available for an extended amount of time. This exit order reinforces the idea that the exit decision exhibits a considerable dynamics and that the monitoring of the investments is of paramount importance for venture capitalists wishing to time divestments.

Second, the achievement of milestones in past rounds accelerates exit by all means. This is in line with the notion that achieving important intermediate results (from a technological perspective) provides better information on the viability of products, whether they are good or bad ones. The degree of value-adding by VC funds also impacts exits. We find that a large syndicate size accelerates exit of all types but the greatest impact is for IPOs. Proximity of at least one VC fund makes trade sales more likely, suggesting that greater control by VC funds is related to more acquisitions. Our third measure of VC value-adding indicates that the VC fund experience makes liquidation more likely. This is suggestive for a more comprehensive screening of investments and possibly higher hurdles required by more established VC firms.

Third, there seems to be little differences in terms of exit routes and timing between portfolio firms located in the Silicon Valley and on Route 128. However, there are significant differences between VC firms located in these two regions and the ones located in other regions of the US. Venture capitalists in Silicon Valley and on Route 128 seem to provide a more favorable environment for entrepreneurial firms, which in turn affects exit outcomes by virtue of being more profitable and having better access to needed resources. Last, we also looked at the impact of stock market conditions on the exit dynamics and found that the exit of investments initiated at times of more favorable IPO conditions tended to be sped up as venture capitalists were probably eager to redirect their human capital towards new investment opportunities.

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